Tensorboard

Tim Book

GENERAL ASSEMBLY

The “Experiment” Problem of Neural Networks

Neural networks have very many tuning parameters. They can also take a very long time to fit. We want a way to…

1. Keep track of our various model runs and their tuning parameters. Each of these model runs can be called an **experiment**.
2. Have an easy-to-navigate UI for in-depth diagnosis of model quality.

What is Tensorboard?

Tensorboard solves this problem! It is a monitoring dashboard for your neural networks.

- It can monitor multiple separate models
- It can monitor them in real-time, as they're fitting
- It provides several relevant visualizations regarding model fit

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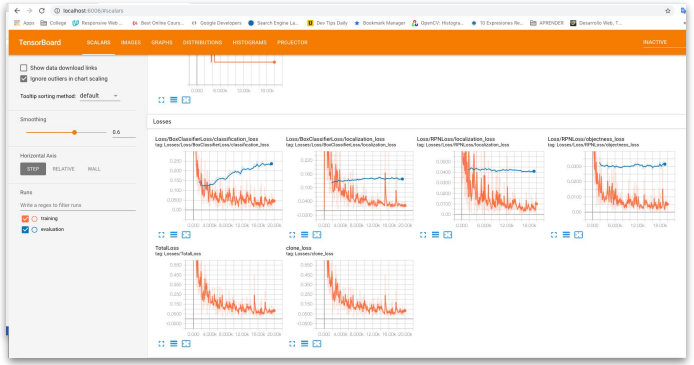
Tensorboard is a “server”

Like many technologies we use, Tensorboard uses the server-client model to communicate with your computer. Here are a few models of how we can use Tensorboard this way:

Tensorboard (while a script is running)

```
$ python script.py
```

Port 6006




```
$ tensorboard --logdir=logs
```

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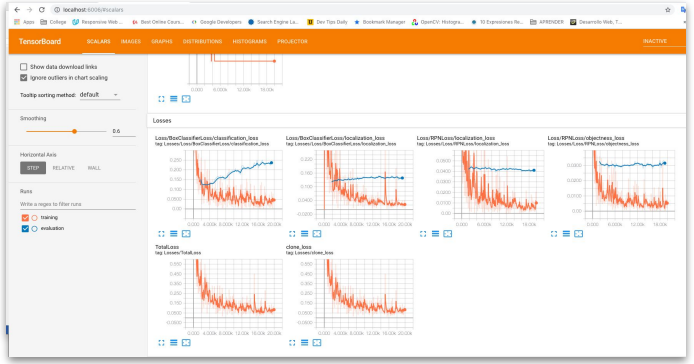
Tensorboard (while Jupyter is running)

Port 8888




```
$ jupyter notebook
```

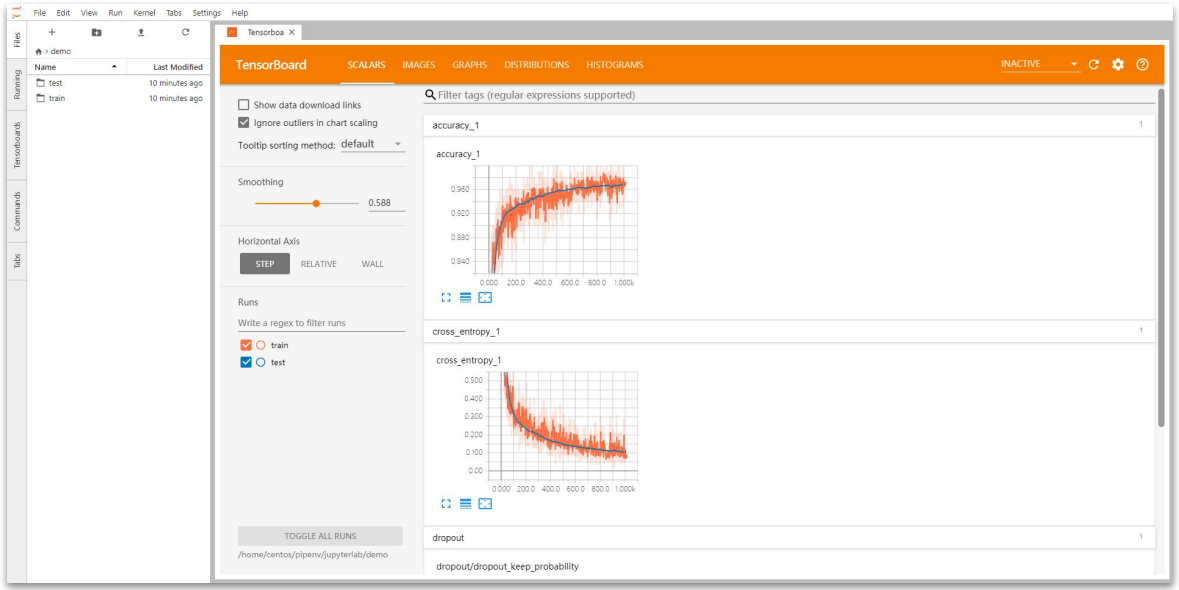
Port 6006




```
$ tensorboard --logdir=logs
```

Tensorboard (inside a notebook!)

Port 8888



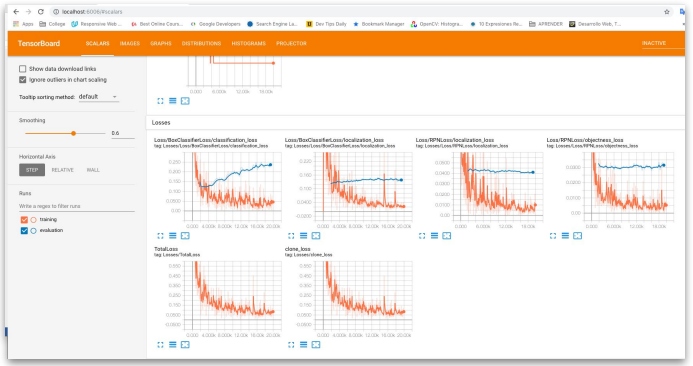

```
%tensorboard --logdir=logs
```

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Tensorboard (from Docker)

Exposing Port 6006

Local Port 6006



Docker Container

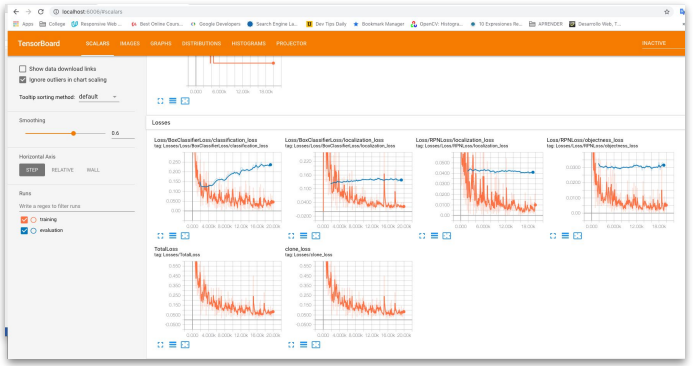
```
$ tensorboard --logdir=logs
```

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Tensorboard (from AWS)

Exposing Port 6006 via SSH

Local Port 6006



AWS EC2 Instance

```
$ tensorboard --logdir=logs
```

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